

Lecture 14

October 26, 2023

Linear Regression Slope Coefficients

Suppose we have de-meaned all variables.

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A more general approach would be helpful.

Define

$$\mathbf{X} = \begin{pmatrix} 1 & X_{11} & X_{21} & \dots & X_{K1} \\ 1 & X_{12} & X_{22} & \dots & X_{K2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} & \dots & X_{Kn} \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbf{e} = \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix},$$

$$\text{and } \hat{\boldsymbol{\beta}} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_K \end{pmatrix}.$$

Then it is equivalent to write the following:

$$Y_1 = \hat{\beta}_0 + \hat{\beta}_1 X_{11} + \hat{\beta}_2 X_{21} + \dots \hat{\beta}_K X_{K1} + \mathbf{e}_1$$

$$Y_2 = \hat{\beta}_0 + \hat{\beta}_1 X_{12} + \hat{\beta}_2 X_{22} + \dots \hat{\beta}_K X_{K2} + \mathbf{e}_2$$

...

$$Y_n = \hat{\beta}_0 + \hat{\beta}_1 X_{1n} + \hat{\beta}_2 X_{2n} + \dots \hat{\beta}_K X_{Kn} + \mathbf{e}_n$$

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Then it is equivalent to write the following:

$$\mathbf{Y} = \mathbf{X}\hat{\boldsymbol{\beta}} + \mathbf{e}.$$

OLS Solution

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Residuals can be written as the following:

$$\begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} - \begin{pmatrix} 1 & X_{11} & X_{21} & \dots & X_{K1} \\ 1 & X_{12} & X_{22} & \dots & X_{K2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} & \dots & X_{Kn} \end{pmatrix} \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_K \end{pmatrix}.$$

$$\mathbf{e} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Minimizing the sum of squared residuals:

$$\hat{\beta} = \operatorname{argmin}_{\mathbf{b}} (\mathbf{Y} - \mathbf{X}\mathbf{b})'(\mathbf{Y} - \mathbf{X}\mathbf{b})$$

Setting derivatives equal to zero and substituting $\hat{\beta}$ for $\mathbf{b}$.

$$2\mathbf{X}^T(\mathbf{Y} - \mathbf{X}\hat{\beta}) = 0$$

$$\mathbf{X}^T\mathbf{e} = 0$$

Let's solve the first equation for $\hat{\beta}$.

Recall that $\mathbf{Y}$ can also be written in terms of β and ϵ

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon.$$

Let's draw a picture to remind ourselves of the difference between $\hat{\beta}$ and β and $\mathbf{e}$ and ϵ .

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2. Bootstrap

Bootstrapping ($\hat{\beta}_k$ for $k = 1, \dots, K$)

1. Take a sample of size n with replacement from your sample
2. Calculate $\hat{\beta}_k^{(b=1)}$ from that bootstrap sample
3. Repeat for $b = 1, \dots, B$ where B is big
4. Do one of the following:
 - ▶ Calculate bootstrapped variance $\frac{1}{B-1} \sum_{b=1}^B (\hat{\beta}_k^{(b)} - \overline{\hat{\beta}_k^{(b)}})^2$
 - ▶ Put $\{\hat{\beta}_k^{(b)} : b = 1, \dots, B\}$ in order and report the $\alpha/2$ and $1 - \alpha/2$ percentiles
 - ▶ ...

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Mean of OLS Estimator

$$\begin{aligned}\widehat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \epsilon) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon\end{aligned}$$

$$\begin{aligned}E[\widehat{\beta}|\mathbf{X}] &= E[\beta|\mathbf{X}] + E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon|\mathbf{X}] \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\epsilon|\mathbf{X}] \\ &= \beta\end{aligned}$$

By iterated expectations,

$$\begin{aligned} E[\hat{\beta}] &= E_{\mathbf{X}}\{E[\hat{\beta}|\mathbf{X}]\} \\ &= E_{\mathbf{X}}\{\beta\} \\ &= \beta \end{aligned}$$

Sampling distributions of the simple and multiple sample slopes
(in the X direction).

i	1	2	...	12		
s	Y_1, X_1, Z_1	Y_2, X_2, Z_2	...	Y_{12}, X_{12}, Z_{12}	$\hat{\beta}_1^s$	$\hat{\beta}_1^m$
1	3.3,97,100	57.2,36,72.6	...	39.3,30,40.6	-.498	-.499
2	30.4,54,99.3	54.0,24,28.5	...	6.2,99,100	-.456	-.398
3	14.7,87,94.8	3.8,98,100	...	3.8,98,100	-.209	-.190
4	:	:	...	:	:	:
5	:	:	...	:	:	:
:	:	:	:	:	:	:
E.(.)	17.3, 70.7,89.3	17.3, 70.7,89.3	...	17.3, 70.7,89.3	-.52	-.33

Law of Total Variance

$$\begin{aligned}V\left[\hat{\beta}\right] &= E\left[V\left[\hat{\beta}|\mathbf{X}\right]\right] + V\left[E\left[\hat{\beta}|\mathbf{X}\right]\right] \\&= E\left[V\left[\hat{\beta}|\mathbf{X}\right]\right] + 0\end{aligned}$$

We can decompose the $\hat{\beta}$ into the expected value and the sampling error.

$$[\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T \mathbf{Y} = \hat{\beta}$$

$$[\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T [\mathbf{X}\beta + \epsilon] = \hat{\beta}$$

$$[\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T \mathbf{X}\beta + [\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T \epsilon = \hat{\beta}$$

$$\beta + [\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T \epsilon = \hat{\beta}$$

Thus:

$$\begin{aligned} E \left[V \left[\hat{\beta} | \mathbf{X} \right] \right] &= E \left[V \left[\beta + \left[\mathbf{X}^T \mathbf{X} \right]^{-1} \mathbf{X}^T \epsilon | \mathbf{X} \right] \right] \\ &= E \left[V \left[\left[\mathbf{X}^T \mathbf{X} \right]^{-1} \mathbf{X}^T \epsilon | \mathbf{X} \right] \right] \\ &= E \left[\left[\mathbf{X}^T \mathbf{X} \right]^{-1} \mathbf{X}^T V[\epsilon | \mathbf{X}] \mathbf{X} \left[\mathbf{X}^T \mathbf{X} \right]^{-1} \right] \end{aligned}$$

Estimation: plug in residuals. This is the “sandwich estimator.”

$$\hat{V} [\hat{\beta}] = [\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X}^T \text{diag} [(\mathbf{Y} - \mathbf{X}\hat{\beta})^2] \mathbf{X} [\mathbf{X}^T \mathbf{X}]^{-1}.$$

This uses the residuals to *estimate* the relationship between the X variables and the errors.

One note: there are many variants of the variance estimators, that involve premultiplying by, e.g., $\frac{n}{n-K}$ or $\frac{n}{n-K-2}$, etc. These are analogous to the $\frac{1}{n-1}$ in the sample variance. It's not so simple with the robust variance estimators (many variants: HC0, HC1, HC2) but all are consistent for the SE.

Many software packages use an alternative. “Classical” or “Homoskedastic” variance estimation. If we *assume* that $V[\epsilon|X_1, X_2, \dots, X_K] = \sigma^2$ for all values of $X_1, X_2, \dots, X_K$: homoskedasticity. Then:

$$\begin{aligned} V[\hat{\beta}] &= \left[E[\mathbf{X}^T \mathbf{X}] \right]^{-1} E \left[\mathbf{X}^T \text{diag}[\epsilon^2] \mathbf{X} \right] \left[E[\mathbf{X}^T \mathbf{X}] \right]^{-1} \\ &= \sigma^2 \left[E[\mathbf{X}^T \mathbf{X}] \right]^{-1} E \left[\mathbf{X}^T \mathbf{I} \mathbf{X} \right] \left[E[\mathbf{X}^T \mathbf{X}] \right]^{-1} \\ &= \sigma^2 \left[E[\mathbf{X}^T \mathbf{X}] \right]^{-1} E \left[\mathbf{X}^T \mathbf{X} \right] \left[E[\mathbf{X}^T \mathbf{X}] \right]^{-1} \\ &= \sigma^2 \left[E[\mathbf{X}^T \mathbf{X}] \right]^{-1} \end{aligned}$$

Where $\sigma^2 = V[\epsilon] = E[\epsilon^2]$.

Classical standard errors make an additional assumption: the magnitude of the errors is *unrelated* to the values of the X s.

This assumption is unlikely to hold. Usually – in real life – the errors will be “heteroskedastic,” because in real life, data is messy. If homoskedasticity *does* hold, then in large samples, the classical and “robust” variance estimates will agree.

With large n and i.i.d. sampling, always use robust standard errors or bootstrapping to characterize the uncertainty of your regression fit.

If Y and X have been centered and x is a nonstochastic scalar (everything works basically the same if we relax non-stochastic assumption)

$$\begin{aligned} V[\hat{\beta}] &= V\left[\beta + \left[\frac{\sum_i x_i \epsilon_i}{\sum_i x_i^2}\right]\right] \\ &= \frac{V[\sum_i x_i \epsilon_i]}{[\sum_i x_i^2]^2} \end{aligned}$$

How does this further simplify if homoskedasticity?

Standard Errors/Variance (with homoskedasticity)

With homoskedasticity, the simple regression model can be written as:

$$y_i = \beta_0^s + \beta_1^s x_i + \epsilon_i^s$$
$$V[\epsilon_i] = \sigma_s^2$$

and the two covariate multiple regression model can be written as:

$$y_i = \beta_0^m + \beta_1^m x_i + \beta_2^m z_i + \epsilon_i^m$$
$$V[\epsilon_i^m] = \sigma_m^2$$

Comparing variances (with homoskedasticity)

$$V[\hat{\beta}_1^s] = \frac{\sigma_s^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$V[\hat{\beta}_1^m] = \frac{1}{1 - R_{x \sim z}^2} \cdot \frac{\sigma_m^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

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If x is randomized, then what does that tell us about

- ▶ β_1^s and β_1^m
- ▶ $V[\hat{\beta}_1^s]$ and $V[\hat{\beta}_1^m]$